

SECURE ACADEMIC CERTIFICATE STORAGE AND VERIFICATION SYSTEM USING BLOCKCHAIN AND SHA-256 HASHING

A PROJECT REPORT

Submitted by

LOGESHWARAN B	611223114050
SRIHARI M	611223114099
PRASANNA J	611223114071
DHAMODHARAN A	611223114018

*In partial fulfillment for the award of the
degree of*

BACHELOR OF ENGINEERING

IN

MECHANICAL ENGINEERING

**KNOWLEDGE INSTITUTE OF TECHNOLOGY
(KIOT Campus, Kakapalayam, Salem – 637504)**

**ANNA UNIVERSITY:CHENNAI 600 025
JUL 2026**

BONAFIDE CERTIFICATE

Certified that this project report “**Secure Academic Certificate Storage and Verification System using blockchain and SHA-256 hashing**” is the Bonafide work of “**LOGESHWARAN. B (61122314050), SRIHARI. M (611223114099), PRASANNA. J (611223114071), DHAMODHARAN. A (611223114018)**” Who carried out the project work under my supervision.

SIGNATURE

**Dr. K.S. PRABHAKARAN, M.E.,
Ph.D.
HEAD OF THE DEPARTMENT
DEPARTMENT OF
MECHANICAL ENGINEERING
KNOWLEDGE INSTITUTE OF
TECHNOLOGY
KIOT CAMPUS,
KAKAPALAYAM (P.O.)
SALEM – 637504**

SIGNATURE

**Prof. A. GAJENDRAN, B.E., M.E.
SUPERVISOR
ASSISTANT PROFESSOR
DEPARTMENT OF
MECHANICAL ENGINEERING
KNOWLEDGE INSTITUTE OF
TECHNOLOGY
KIOT CAMPUS,
KAKAPALAYAM (P.O.)
SALEM – 637504**

ABSTRACT

The Blockchain-Based Academic Certificate Verification System is a web-based application developed to provide a secure, reliable, and efficient method for issuing, storing, and verifying academic certificates. Traditional certificate verification methods generally depend on manual document checking and institutional communication, which can be time-consuming and may be vulnerable to document forgery or unauthorized modification. The proposed system addresses these limitations by integrating blockchain technology, SHA-256 hashing, QR-code-based verification, and a digital certificate management system.

In the proposed system, an authorized administrator can upload and store academic certificate details along with the corresponding certificate file. A unique certificate ID is assigned to each certificate, and a SHA-256 hash is generated from the uploaded certificate to maintain its digital integrity. The certificate information is organized into blockchain records, where each block contains the hash of the previous block, creating a chain of interconnected records. This structure helps in detecting unauthorized modification of certificate information.

A QR code is generated for every certificate and contains a unique verification URL. When the QR code is scanned using a mobile device, the user is directed to the certificate verification page, where the certificate details can be viewed and the original certificate can be downloaded. The system also provides features such as certificate searching, certificate management, blockchain integrity verification, QR-code generation, certificate deletion, and secure certificate downloading.

The application is developed using Python Flask, SQLite, HTML, CSS, SHA-256 hashing, QR-code generation, and blockchain-based data management. The system can be deployed on a cloud platform, enabling certificate verification through Internet-connected devices. The proposed system reduces manual verification effort, improves certificate accessibility, strengthens data integrity, and provides a practical digital approach for reliable academic certificate verification.

TABLE OF CONTENT

CHAPTER NO.	DESCRIPTION	PAGE NO.
	ABSTRACT	i
	LIST OF FIGURES	iv
	LIST OF TABLES	v
1	INTRODUCTION	1
	1.1 Methodology	2
2	LITERATURE REVIEW	4
3	SYSTEM ANALYSIS	6
	3.1 Existing System	6
	3.2 Proposed System	7
	3.3 Advantages of Proposed System	8
	3.4 Functional Requirements	9
	3.5 Non-Functional Requirements	11
	3.6 Feasibility Analysis	12
4	SYSTEM DESIGN	14
	4.1 System Architecture	14
	4.2 Project Workflow	15
	4.3 Use Case Diagram	16
	4.4 Activity Diagram	18
	4.5 Sequence Diagram	18
	4.6 Database Design	19
5	SYSTEM IMPLEMENTATION	21
	5.1 Development Environment	21
	5.2 Admin Authentication Module	21
	5.3 Certificate Upload and Storage Module	22
	5.4 Certificate ID Management	22
	5.5 SHA-256 Hash Generation	23
	5.6 Blockchain Record Generation	23
	5.7 Blockchain Integrity Verification	24

CHAPTER NO.	DESCRIPTION	PAGE NO.
	5.8 QR Code Generation	24
	5.9 Certificate Verification Module	25
	5.10 Certificate Search Module	25
	5.11 Certificate Management Module	26
	5.12 Certificate Download Module	26
6	RESULTS AND DISCUSSION	27
	6.1 Home Page	27
	6.2 Admin Login Page	27
	6.3 Dashboard	28
	6.4 Certificate Upload Result	29
	6.5 Certificate Management Result	30
	6.6 Certificate Search Result	31
	6.7 Certificate Verification Result	32
	6.8 QR Code Generation Result	33
	6.9 Blockchain Explorer Result	34
	6.10 Blockchain Integrity Result	34
	6.11 Certificate Download Result	35
7	CONCLUSION AND FUTURE SCOPE	36
	7.1 Conclusion	36
	7.2 Future Scope	37
8	APPENDICES	39
	8.1 Pseudocode	39
	8.2 HTML Templates	44
	8.3 CSS Stylesheet	45
	8.4 Database Schema	46
9	REFERENCES	48

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
1.1	Working Methodology	3
4.1	System Architecture Diagram	15
4.2	Use Case Diagram	17
4.3	Activity Diagram	18
4.4	Sequence Diagram	19
6.1	Home Page	27
6.2	Admin Login Page	28
6.3	Dashboard	29
6.4	Certificate Upload	30
6.5	Certificate Management	31
6.6	Certificate Search	32
6.7	Certificate Verification	33
6.8	QR Code Generation	34
6.9	Blockchain Explorer	34
6.10	Certificate Download	35

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
4.1	Use case Relationships	17
4.2	Certificate Table	20
4.3	Admin Table	20
8.1	Certificate Database Schema	46
8.2	Admin Database Schema	47

CHAPTER 1

INTRODUCTION

Academic certificates are important documents used to prove the educational qualifications and achievements of students. Certificates are commonly required for higher education, employment, professional applications, and other official purposes. However, conventional certificate verification methods generally depend on manual checking of physical documents or contacting the issuing institution. These methods can be time-consuming and may make it difficult to verify certificates quickly and reliably.

The Blockchain-Based Academic Certificate Verification System is proposed as a web-based solution for secure certificate management and verification. The system allows an authorized administrator to upload certificate information and the corresponding certificate file. Each certificate is assigned a unique certificate ID and stored in a structured database. The uploaded certificate is processed using the SHA-256 hashing algorithm to generate a unique digital hash that helps maintain the integrity of the certificate.

Blockchain technology is incorporated to maintain certificate records in a chain of blocks. Each certificate record contains the hash of the previous record along with its own block hash. This creates a linked structure that helps identify unauthorized modifications to stored certificate information.

The system also generates a QR code for each certificate. The QR code is intended for students, users, or other authorized persons who need to verify a certificate. By scanning the QR code, the user can access the certificate verification page and view the stored certificate details. The system also provides an option to download the corresponding certificate file.

The proposed system therefore combines web technology, database management, SHA-256 hashing, blockchain concepts, and QR-code-based verification to provide a reliable and convenient approach for academic certificate verification.

1.1 METHODOLOGY

The proposed Blockchain-Based Academic Certificate Verification System is developed as a web-based application to provide a secure and efficient method for managing and verifying academic certificates. The methodology begins with administrator authentication, where the authorized administrator logs into the system before accessing certificate management functions. After successful authentication, the administrator enters the required certificate information such as certificate ID, student name, register number, course, institution, issue date, and certificate file. The uploaded certificate is validated and stored securely, while the corresponding certificate information is maintained in the SQLite database.

After the certificate is uploaded, the system generates a SHA-256 hash from the certificate file. This hash acts as a digital fingerprint and helps maintain the integrity of the uploaded document. The certificate information, hash value, previous block hash, and other relevant details are then used to create a blockchain record. Each newly created block is connected with the previous block through its hash value, forming a sequential chain of certificate records. The system can verify the integrity of this chain by recalculating the block hashes and comparing them with the stored values.

For convenient certificate verification, the system generates a unique QR code for every certificate. The QR code contains the verification URL associated with the certificate. Students or users can scan the QR code to access the certificate verification page. The system retrieves the corresponding certificate record from the database and displays important information such as the certificate ID, student name, register number, course, institution, and issue date. The user can also access and download the corresponding certificate file from the verification page.

The system further provides certificate management and search facilities for the administrator. Previously stored certificates can be viewed and searched using relevant certificate information. The blockchain explorer provides access to the stored blockchain records and allows the administrator to examine the sequence and integrity of certificate blocks. Through the combination of Flask, SQLite,

SHA-256 hashing, blockchain-based record management, and QR-code technology, the proposed methodology provides a structured approach for certificate storage, integrity protection, and convenient verification. This reduces the dependency on manual verification procedures and provides a reliable digital method for managing and verifying academic certificates.

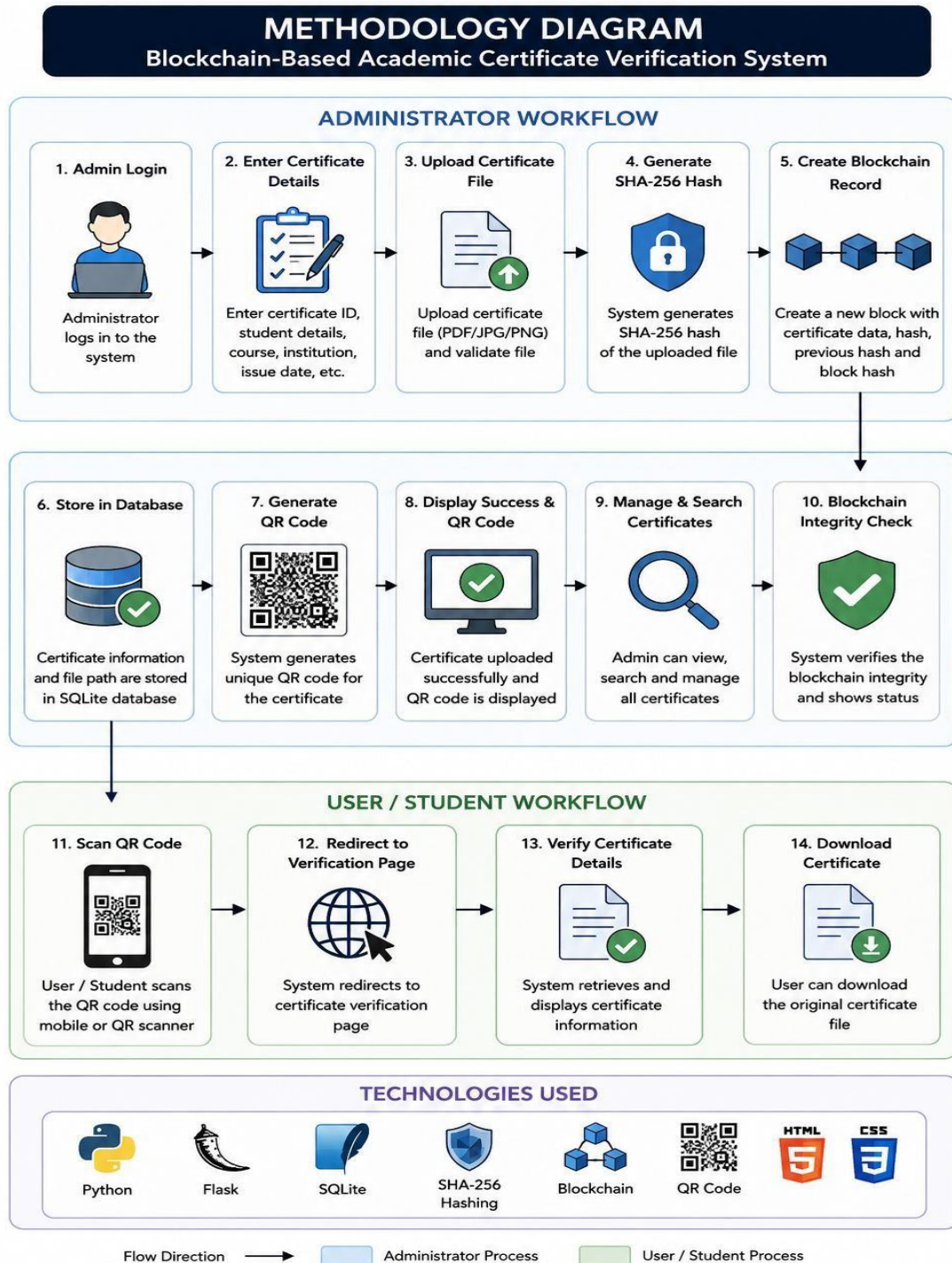


Figure 1.1 Working Methodology

CHAPTER 2

LITERATURE REVIEW

Sharples and J. Domingue (2016)

M. Sharples and J. Domingue (2016) suggested the application of blockchain technology for storing educational records in a convenient way. The article described the opportunity of using it for keeping students' achievements in order that they cannot be changed or falsified. The authors also discussed how this technology will guarantee the student's knowledge and provide transparency. Their main point was that blockchain can eliminate the single source of authority over the data and give students an opportunity to save their achievements in a safe place and share them without any restrictions. Therefore, this article can be used in the chosen field for suggesting ways to design certificate verification technology based on blockchain.

Dennis Yue, Joseph T. et al. (2018)

Dennis Yue with his team of researchers proposed a blockchain-based framework for issuing and verifying academic certificates in 2018. They designed a mechanism that allows storing information about the certificate and its hash in blockchain technology, thus ensuring the impossibility of forgery. The study showed that employers and institutions can quickly verify certificates without contacting the issuing department. In addition, the researchers discussed reducing the burden of administrative work and increasing the reliability of verifying academic documents. All this led to the implementation of blockchain technology in the document management system of education.

Zibin Zheng et al. (2018)

Zibin Zheng et al. (2018) provide a survey of blockchain technology that covers its architecture, consensus mechanisms, security, and applications. The paper gives an insight into the inner workings of blockchain technology, explaining how it utilizes hashing and linking of blocks to secure information and data.

The authors emphasize the technology's ability to offer data protection and transparency. The study also revealed education as one of the fields that can benefit from the implementation of blockchain technology. Specifically, the paper provides a valuable source of information for designing certificate verification based on blockchain.

Satoshi Nakamoto (2008)

Satoshi Nakamoto (2008) presented the concept of blockchain, which is applied in the Bitcoin cryptocurrency. The paper describes the process of recording transactions in blocks linked by hash functions to ensure authenticity. While the article focuses on implementing the technology in digital money, it can be used for other purposes, such as storing academic certificates. The concepts presented in the paper can be the basis of certificate verification systems since they describe the blockchain technology used nowadays to store and secure data.

NIST - Lily Chen (2015)

Lily Chen along with National Institute of Standards and Technology developed the Secure Hash Algorithm (SHA-256). The latter is considered to be an effective cryptographic hashing method due to which it becomes possible to ensure the data's integrity. According to the researchers, SHA-256 relies on a unique 256-bit hash value that makes it possible to eliminate the possibility of having two different files with the same hash. They also mentioned that any alteration of the document leads to the significant difference of its hash. In addition to the possibility for detecting unauthorized changes, this method also is applied when blockchain technologies are used, which is the main security guarantee for certificate verification applications.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing certificate management and verification process in many educational institutions is largely dependent on manual documentation and conventional digital storage methods. Certificates are generally created, stored, and verified using paper-based records, local files, or centralized databases. When a student or other authorized person needs to verify a certificate, the certificate details may have to be manually checked against institutional records. This process can consume considerable time and may require the involvement of administrative personnel.

In conventional digital certificate management systems, certificate information is commonly stored in a database without maintaining a strong relationship between individual certificate records. If the stored certificate file is modified or replaced, identifying such changes can be difficult. Similarly, maintaining the authenticity and integrity of certificates becomes challenging when verification depends only on the information printed on the certificate.

Another limitation of traditional verification is the absence of a quick mechanism for accessing certificate information. Users may need to manually enter certificate numbers or contact the institution for confirmation. This can make the verification process inconvenient, particularly when a large number of certificates are issued and maintained.

Conventional systems may also lack a transparent record of certificate-related data. Searching for a particular certificate based on certificate ID, student name, or register number can become difficult when records increase over time. Manual record maintenance may also lead to duplication, loss of information, and human errors.

Therefore, there is a need for a computerized certificate management system that can securely store certificate records, maintain their integrity, provide efficient searching, and allow users to verify certificate information easily.

3.2 PROPOSED SYSTEM

The proposed system is a Blockchain-Based Academic Certificate Verification System developed to improve the storage, management, and verification of academic certificates. The system combines a web-based certificate management interface, SQLite database, SHA-256 hashing, blockchain-based record linking, and QR code technology.

The system allows authorized users to upload academic certificates along with important details such as certificate ID, student name, register number, course, institution, issue date, and certificate file. The uploaded certificate file is stored securely, while a SHA-256 hash is generated from the file. This hash acts as a digital fingerprint of the certificate and helps in maintaining its integrity.

When a certificate is uploaded, a blockchain record is created containing the certificate information, certificate hash, previous block hash, timestamp, and current block hash. The previous hash connects the current certificate record with the preceding record, forming a sequential chain of certificate records. Any unexpected modification in the stored blockchain information can therefore be detected during integrity verification.

The system also generates a unique QR code for each certificate. The QR code contains a verification link associated with the certificate record. A student, employer, institution, or other authorized user can scan the QR code using a mobile device and access the certificate verification page. The verification page displays the stored certificate information, allowing the user to compare the details with the original certificate.

The system additionally provides a Manage Certificates module through which certificate records can be viewed and searched. Certificates can be searched using certificate ID, student name, or register number. The system provides a certificate download facility so that authorized users can retrieve the uploaded certificate file when required.

The proposed system therefore provides an integrated approach for certificate storage, integrity protection, searching, QR-based verification, and blockchain record management.

Main features of the proposed system

- Secure certificate upload and storage
- Certificate ID-based identification
- SHA-256 certificate hashing
- Blockchain-based certificate record creation
- Previous-hash and current-hash linking
- QR code generation for certificate verification
- QR-based access to certificate details
- Certificate search facility
- Certificate download facility
- Blockchain integrity verification
- Certificate management through a web interface

3.3 ADVANTAGES OF PROPOSED SYSTEM

The proposed system provides several advantages over conventional certificate management and verification methods.

1. Improved Certificate Integrity

SHA-256 hashing generates a unique digital fingerprint for each uploaded certificate file. This makes it possible to identify changes made to the stored certificate.

2. Blockchain-Based Record Linking

Each certificate record contains the hash of the previous record. This creates a sequential blockchain structure and improves the traceability of certificate records.

3. Quick QR Code Verification

A unique QR code is generated for each certificate. Users can scan the QR code to quickly access the corresponding certificate verification page instead of manually searching for the certificate.

4. Reduced Manual Verification

The system reduces the need for manual checking of certificate records. Certificate information can be retrieved directly from the system.

5. Easy Certificate Searching

The Manage Certificates module allows certificates to be searched using important details such as certificate ID, student name, and register number.

6. Centralized Digital Management

Certificate information and uploaded files can be managed through a single web-based system, making record management easier.

7. Easy Certificate Retrieval

The download facility allows authorized users to retrieve stored certificate files when required.

8. Integrity Verification

The blockchain integrity checking mechanism verifies the relationship between certificate records and detects inconsistencies in the blockchain structure.

9. Better Record Organization

Certificates are maintained systematically with details including certificate ID, student information, course, institution, issue date, file name, hash values, and timestamps.

10. User-Friendly Verification

QR-based verification simplifies the process for students and other users because certificate details can be accessed directly through the QR code.

3.4 FUNCTIONAL REQUIREMENTS

Functional requirements define the operations that the proposed system must perform.

1. User Authentication

The system shall provide a login mechanism for authorized users. The system shall verify the entered username and password before providing access to certificate management functions.

2. Certificate Upload

The system shall allow authorized users to upload certificate files in supported formats such as PDF, PNG, JPG, and JPEG. Certificate information such as certificate ID, student name, register number, course, institution, and issue date shall also be entered.

3. Certificate Storage

The system shall store uploaded certificate files and their corresponding information in the system. Each certificate shall be associated with a unique certificate ID.

4. SHA-256 Hash Generation

The system shall generate a SHA-256 hash from the uploaded certificate file. The generated hash shall be stored along with the certificate record.

5. Blockchain Record Creation

The system shall create a blockchain record for every uploaded certificate. The record shall contain certificate information, certificate hash, previous hash, current block hash, and timestamp.

6. QR Code Generation

The system shall automatically generate a unique QR code for each uploaded certificate. The QR code shall contain the verification URL associated with the certificate.

7. Certificate Verification

The system shall provide a certificate verification facility. When the QR code is scanned, the corresponding certificate details shall be retrieved and displayed through the verification page.

8. Certificate Search

The system shall allow users to search stored certificates using certificate ID, student name, or register number.

9. Certificate Management

The system shall display stored certificates in the Manage Certificates module along with their important details and available actions.

10. Certificate Download

The system shall allow the stored certificate file to be downloaded through the certificate management or verification interface.

11. Blockchain Integrity Verification

The system shall verify the blockchain records by recalculating block hashes and comparing them with the stored hash values.

12. Certificate Display

The system shall display certificate details including:

- Certificate ID
- Student Name
- Register Number
- Course
- Institution
- Issue Date
- Certificate status

3.5 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality and performance characteristics of the system.

1. Security

The system should protect certificate information and restrict certificate management functions to authorized users.

2. Integrity

Certificate records should maintain their original hash values and blockchain relationships so that unauthorized modifications can be detected.

3. Reliability

The system should provide consistent results when uploading, storing, searching, and verifying certificates.

4. Usability

The web interface should be simple and easy to understand. Users should be able to navigate between certificate upload, management, search, verification, and blockchain modules without difficulty.

5. Performance

Certificate information should be retrieved within a reasonable time. Searching and verification operations should respond efficiently even when multiple certificate records are stored.

6. Scalability

The system should be capable of handling an increasing number of certificate records without requiring major changes to its basic architecture.

7. Maintainability

The system should be developed using modular components such as Flask routes, database functions, blockchain functions, QR code generation, and HTML templates so that individual components can be modified easily.

8. Availability

The system should be accessible whenever the application and its supporting database are running, allowing authorized users to perform certificate management and verification operations.

9. Compatibility

The system should work with commonly used web browsers and support certificate files in standard formats such as PDF, PNG, JPG, and JPEG.

10. Accuracy

The system should accurately retrieve certificate information using the certificate ID and QR verification link. Hash and blockchain integrity calculations should produce consistent results.

3.6 FEASIBILITY ANALYSIS

Feasibility analysis determines whether the proposed Blockchain-Based Academic Certificate Verification System can be practically developed and used.

1. Technical Feasibility

The system is technically feasible because it can be developed using commonly available software technologies. Python and Flask are used for web application development, while SQLite is used for database management. SHA-256 hashing can be implemented using Python libraries, and QR codes can be generated using a Python QR code library.

The system does not require highly specialized hardware. It can operate using a standard computer with Python, a web browser, and the required software libraries. Therefore, the proposed system is technically feasible.

2. Economic Feasibility

The proposed system is economically feasible because it primarily uses software technologies and open-source tools. Python, Flask, SQLite, HTML, CSS, and the required Python libraries can be used without significant software licensing costs. The system can therefore be developed with comparatively low cost while providing useful certificate management and verification features.

3. Operational Feasibility

The system is operationally feasible because its interface is designed to be simple and user-friendly. Authorized users can upload certificates, manage records, search certificates, view blockchain records, and download certificates through the web interface.

The QR code provides an easy method for users to access certificate verification details. Therefore, the proposed system can be practically operated without requiring advanced technical knowledge.

4. Schedule Feasibility

The system can be developed within an academic project schedule because it uses well-established technologies and modular development. The project can be divided into stages such as database development, certificate upload, hashing, blockchain implementation, QR generation, verification, searching, and testing.

5. Overall Feasibility

Based on technical, economic, operational, and schedule considerations, the proposed system is feasible for academic certificate management and verification.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The Blockchain-Based Academic Certificate Verification System is designed as a web-based application that provides a secure and reliable method for issuing, storing, managing, and verifying academic certificates. The system follows a layered architecture in which the user interacts with the web interface, while the Flask application handles the application logic and communicates with the SQLite database.

The system consists of the following major layers:

- ***User Interface Layer*** – Provides pages for login, dashboard, certificate upload, certificate management, certificate search, verification, and QR-code-based verification.
- ***Application Layer*** – Developed using Python Flask, this layer processes user requests, validates certificate information, generates hashes, creates QR codes, and manages verification operations.
- ***Security and Blockchain Layer*** – SHA-256 is used to generate a unique hash for the uploaded certificate. Each certificate record is linked with the previous block hash to maintain the blockchain structure.
- ***Database Layer*** – SQLite stores certificate details, uploaded file information, QR-code information, certificate hashes, previous hashes, block hashes, and timestamps.
- ***File Storage Layer*** – Uploaded certificate files and generated QR-code images are stored in their respective folders.

4.1 SYSTEM ARCHITECTURE

The system is a web-based application that allows administrators to upload academic certificates and manage them securely. Students/Users can scan QR codes to verify certificate authenticity and view certificate details.

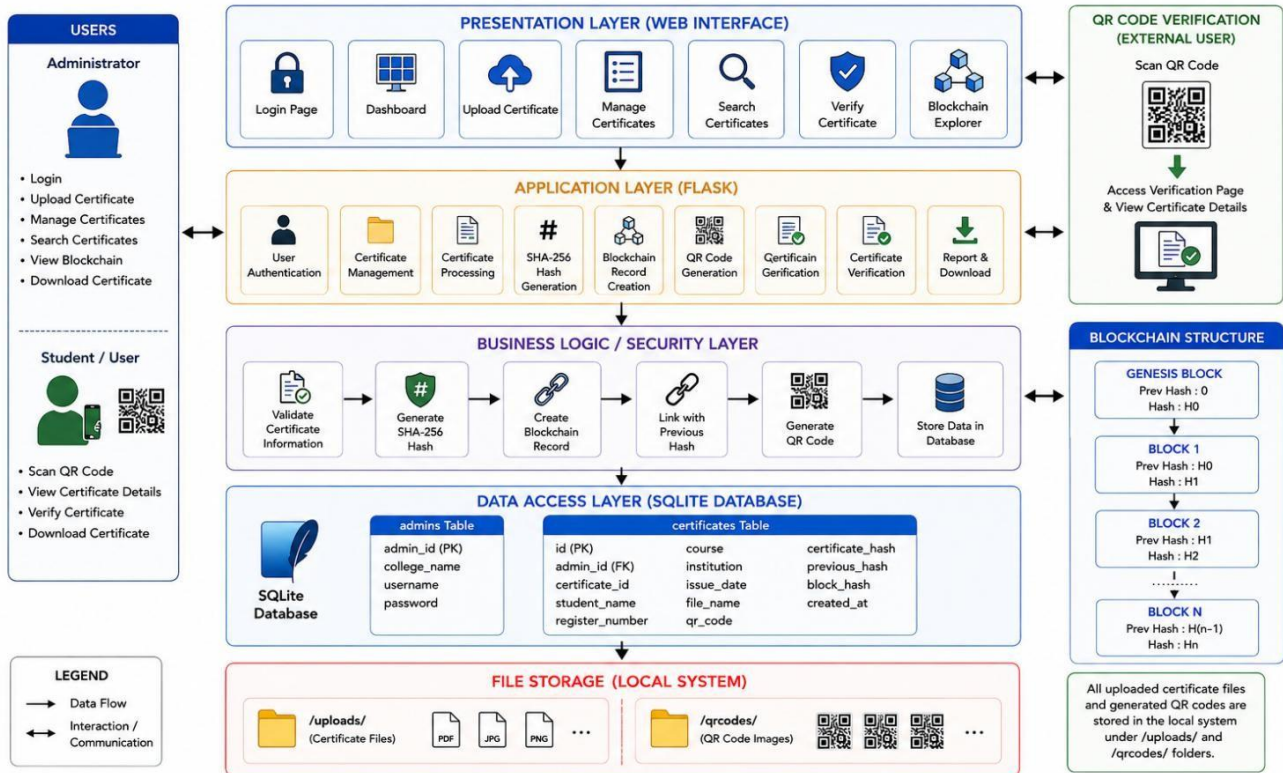


Figure 4.1 System Architecture

4.2 PROJECT WORKFLOW

The project workflow describes the complete sequence of operations performed by the system from certificate uploading to certificate verification.

Certificate Upload Workflow

- The administrator logs into the system.
- The administrator opens the Certificate Upload module.
- Certificate details such as Certificate ID, Student Name, Register Number, Course, Institution, and Issue Date are entered.
- The certificate file is selected and uploaded.
- The system validates the entered information and file type.
- A SHA-256 hash is generated for the uploaded certificate file.
- The system obtains the hash of the previous certificate record.
- A new blockchain record is created using the certificate information, certificate hash, previous hash, and timestamp.

- A QR code is generated containing the certificate verification URL.
- Certificate details, blockchain information, and QR-code information are stored in the SQLite database.
- The certificate becomes available in the Manage Certificates module.

Certificate Verification Workflow

- The user/student scans the QR code printed or associated with the certificate.
- The QR code opens the certificate verification URL.
- The Flask application receives the Certificate ID and administrator identifier.
- The system searches the SQLite database for the corresponding certificate.
- If the certificate exists, its details are retrieved.
- The certificate details are displayed to the user.
- The displayed information can be used to confirm the authenticity of the certificate.

4.3 USE CASE DIAGRAM

The use case diagram represents the interactions between the system users and the major functions provided by the application.

Main Actor

Administrator

- Login
- Upload Certificate
- View Certificates
- Search Certificates
- Verify Certificate
- View Blockchain Records
- Download Certificate

User/Student

- Scan QR Code
- View Certificate Details
- Verify Certificate Information

4.3 USE CASE DIAGRAM

The use case diagram represents the interactions between the actors (Administrator and Student/User) and the major functions provided by the system.

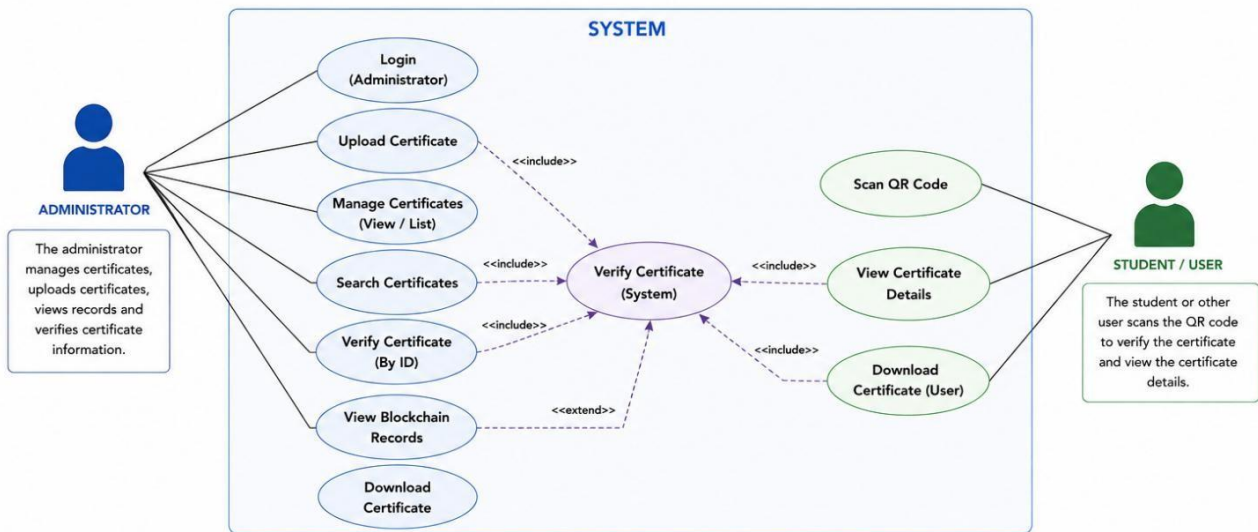


Figure 4.2 Usecase Diagram

Use Case Relationships

Actor	Use Case
Administrator	Login
Administrator	Upload Certificate
Administrator	Manage Certificates
Administrator	Search Certificate
Administrator	Verify Certificate
Administrator	View Blockchain Records
Administrator	Download Certificate
User/Student	Scan QR Code
User/Student	View Certificate Details

Table 4.1 Use case Relationships

4.4 ACTIVITY DIAGRAM

The activity diagram represents the step-by-step activities involved in certificate processing and verification.

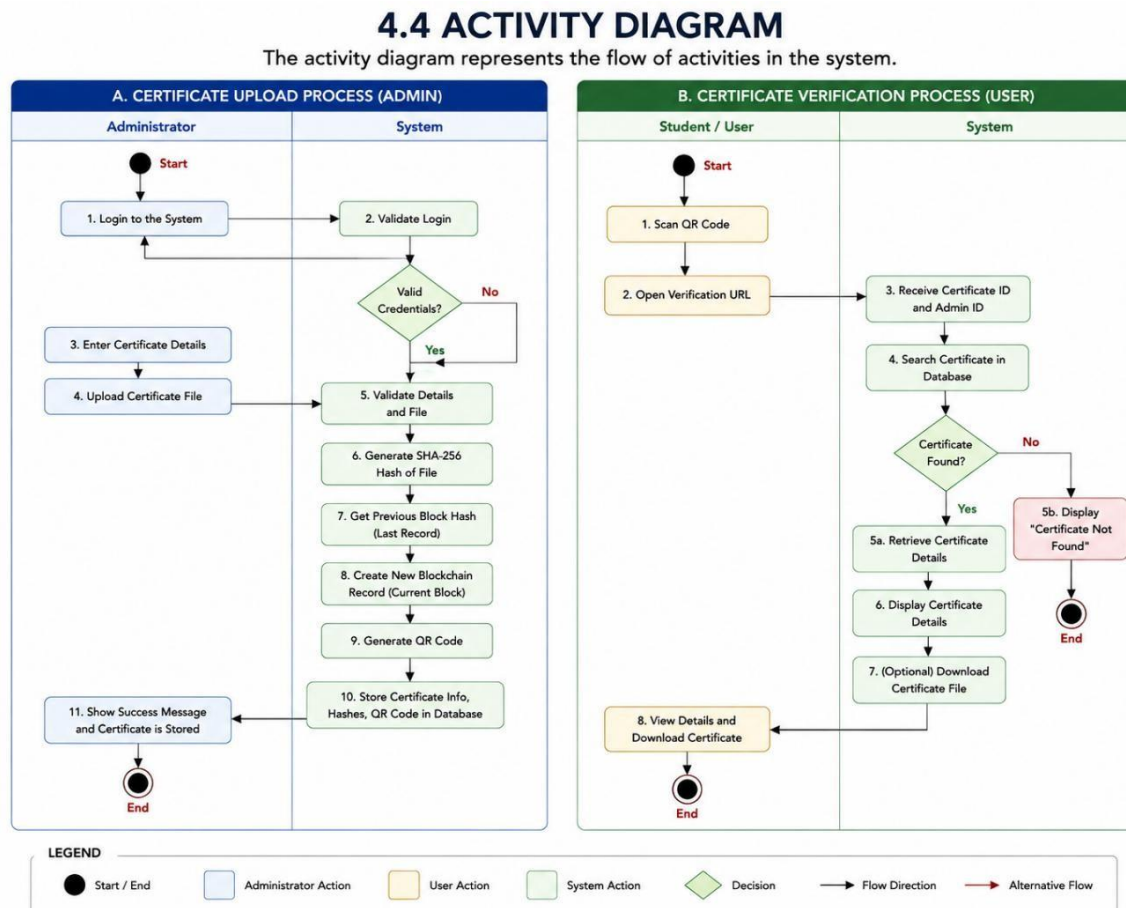


Figure 4.3 Activity Diagram

4.5 SEQUENCE DIAGRAM

The sequence diagram describes the interaction between the user, web application, database, blockchain module, and QR-code generator.

Certificate Upload Sequence

- Administrator sends login credentials to the Flask application.
- Flask validates the credentials with the database.
- Administrator submits certificate details and certificate file.
- Flask receives the certificate.
- Flask generates the SHA-256 certificate hash.
- Flask obtains the previous block hash from the database.

- Blockchain module generates the new block hash.
- QR-code generator creates the QR code containing the verification URL.
- Flask stores the certificate information in SQLite.
- Flask displays the successful upload result.

QR Code Verification Sequence

- User/Student scans the QR code.
- QR code opens the verification URL.
- Flask receives the administrator ID and Certificate ID.
- Flask sends a database query.
- SQLite searches for the matching certificate.
- Database returns the certificate information.
- Flask displays the certificate details.
- User/Student verifies the displayed information.

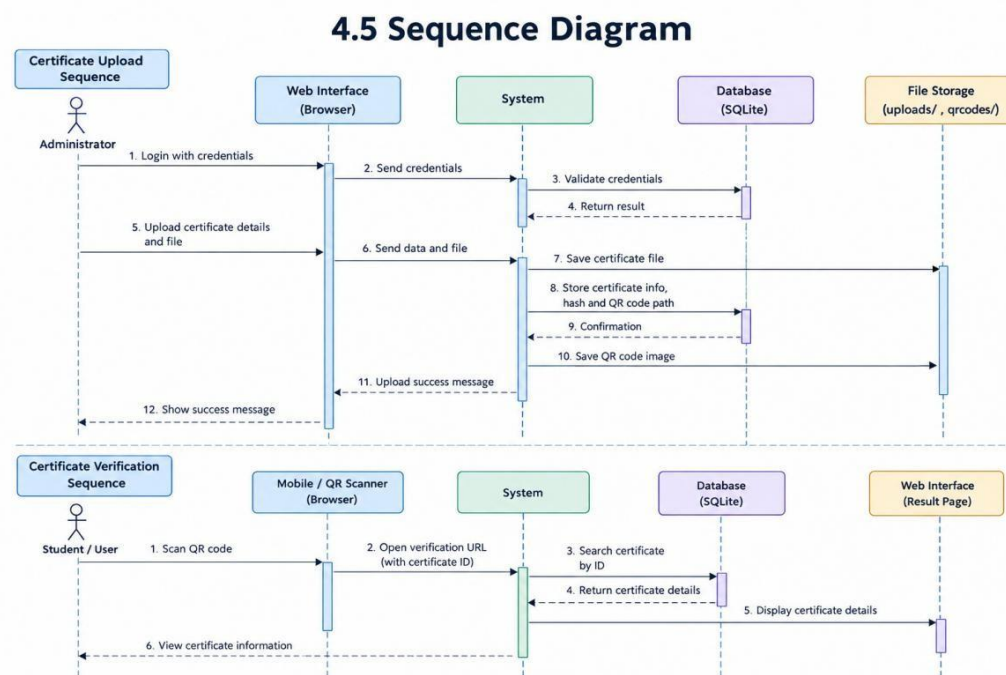


Figure 4.4 Sequence Diagram

4.6 DATABASE DESIGN

The system uses SQLite as the database management system. The database stores administrator information and certificate records.

Certificate Table

The main certificates table contains the following fields:

Field Name	Data Type	Description
id	INTEGER	Unique database record number
admin_id	INTEGER	Identifies the administrator associated with the certificate
certificate_id	TEXT	Unique certificate identification number
student_name	TEXT	Name of the student
register_number	TEXT	Student register number
course	TEXT	Academic course
institution	TEXT	Institution name
issue_date	TEXT	Certificate issue date
file_name	TEXT	Uploaded certificate file name
qr_code	TEXT	Generated QR-code file name
certificate_hash	TEXT	SHA-256 hash of the certificate
previous_hash	TEXT	Hash of the previous blockchain record
block_hash	TEXT	Hash of the current blockchain record
created_at	TEXT	Date and time of record creation

Table 4.2 Certificate Table

Admin Table

The admins table stores administrator login information.

Field Name	Data Type	Description
admin_id	INTEGER	Unique administrator ID
college_name	TEXT	Institution associated with the administrator
username	TEXT	Login username
password	TEXT	Login password

Table 4.3 Admin Table

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 DEVELOPMENT ENVIRONMENT

The development environment provides the necessary software tools and technologies required to design, develop, test, and execute the certificate verification system. The project is developed using **Python** as the primary programming language because of its simplicity, flexibility, and availability of libraries for web development, hashing, database management, and QR code generation. **Flask** is used as the web application framework to create the different pages and handle user requests. **SQLite** is used as the database because it is lightweight and suitable for storing certificate and administrator information. **Visual Studio Code** is used as the integrated development environment for writing and managing the source code. HTML and CSS are used to design the web pages and provide a user-friendly interface. The system also uses a local file storage folder for uploaded certificate files and QR code images. These technologies work together to provide a simple and efficient certificate management and verification platform.

5.2 ADMIN AUTHENTICATION MODULE

The Admin Authentication Module provides controlled access to the certificate management functions of the system. The administrator must enter a valid username and password through the login page before accessing the dashboard. The submitted login information is checked against the administrator records stored in the SQLite database. When the credentials are correct, a secure session is created and the administrator is redirected to the dashboard. The session also identifies the administrator associated with the certificates being managed. This prevents unauthorized users from accessing certificate upload, certificate management, blockchain records, and other administrative functions. If incorrect credentials are entered, an appropriate error message is displayed and access to the dashboard is denied. A logout function is also provided to terminate the active session. Thus, the

authentication module provides a basic security layer for protecting certificate records and administrative operations within the system.

5.3 CERTIFICATE UPLOAD AND STORAGE MODULE

The Certificate Upload and Storage Module is responsible for adding new academic certificate records to the system. The administrator enters important certificate information such as Certificate ID, student name, register number, course, institution, and issue date. The administrator then selects the corresponding certificate file from the computer. The system validates the entered information and checks whether the uploaded file belongs to an allowed format such as PDF, PNG, JPG, or JPEG. After successful validation, the certificate file is stored in the project's upload folder with a suitable file name. The certificate information is stored in the SQLite database along with the file name and other verification details. During the same process, a SHA-256 hash is generated for the uploaded certificate and a QR code is created for verification. This module provides an organized method for storing certificate files and their associated digital information.

5.4 CERTIFICATE ID MANAGEMENT

The Certificate ID Management module provides a unique identification mechanism for every certificate stored in the system. Each certificate is assigned a Certificate ID during the upload process. The Certificate ID is used to identify and retrieve a particular certificate from the database during verification and certificate management operations. The system maintains the relationship between the Certificate ID, student details, uploaded file, QR code, hash value, and blockchain record. The database also prevents duplicate Certificate IDs for the same administrator, ensuring that two certificates cannot unintentionally use the same identifier. In the Manage Certificates page, certificates are displayed with a sequential display number based on their current order. This display number helps administrators easily identify certificate records in the table. The actual certificate information remains associated with its stored

database record. Therefore, Certificate ID Management provides an organized and reliable method for identifying certificates throughout the system.

5.5 SHA-256 HASH GENERATION

The SHA-256 Hash Generation module provides a digital fingerprint for every uploaded certificate file. When a certificate is uploaded, the system reads the contents of the file and generates a SHA-256 hash value. A hash is a fixed-length digital value generated from the contents of a file. Even a small change in the certificate file can produce a different hash value. The generated hash is stored in the database along with the certificate information and is also used during blockchain record generation. This provides an additional method of protecting the integrity of certificate records. The hash does not store the original certificate contents; instead, it represents the certificate file using a unique digital fingerprint. By associating the SHA-256 hash with the corresponding certificate record, the system can identify whether the stored certificate data has remained consistent. Thus, SHA-256 plays an important role in maintaining the integrity and reliability of the certificate verification system.

5.6 BLOCKCHAIN RECORD GENERATION

The Blockchain Record Generation module stores certificate information in a sequential blockchain-style structure. Whenever a new certificate is uploaded, the system creates a record containing important certificate information, the SHA-256 certificate hash, a timestamp, and the hash of the previous certificate record. The previous hash establishes a connection between the current record and the preceding record. A new block hash is then generated using the relevant certificate information and previous hash. This creates a chain of linked records in which each record depends on the information stored in the previous record. The blockchain records are stored in the SQLite database and can be viewed through the Blockchain Explorer section of the system. This approach provides a transparent record of issued certificates and makes unauthorized modification easier to identify. The blockchain concept therefore improves the integrity and traceability of certificate records.

5.7 BLOCKCHAIN INTEGRITY VERIFICATION

The Blockchain Integrity Verification module checks whether the stored certificate records maintain their original blockchain relationships. The system retrieves the certificate records in their stored order and begins the verification process from the first block. The first record is expected to have the initial previous hash value. For every subsequent record, the system compares its stored previous hash with the block hash of the preceding record. The system also recalculates the block hash using the certificate information, certificate hash, previous hash, and timestamp. The recalculated value is compared with the stored block hash. If all comparisons are successful, the blockchain is considered valid. If any value has been modified or the chain relationship has been disturbed, the integrity verification process reports that the chain is not valid. This module helps demonstrate that the certificate records have maintained their expected blockchain structure.

5.8 QR CODE GENERATION

The QR Code Generation module provides a convenient method for accessing certificate verification information. When a certificate is successfully uploaded, the system generates a unique QR code associated with that certificate. The QR code contains a verification URL that identifies the administrator and the corresponding Certificate ID. The generated QR code image is stored in the project's QR code storage folder, while its file name is stored in the database. The QR code is displayed in the Manage Certificates page along with the related certificate record. A student, user, or authorized person can scan the QR code using a QR scanning application to access the certificate verification page. The verification page displays the relevant certificate details when the record is available. This reduces the need for manually entering certificate information and provides a simple method for accessing certificate verification details. Thus, QR codes improve the usability of the verification system.

5.9 CERTIFICATE VERIFICATION MODULE

The Certificate Verification Module is one of the main components of the system and is responsible for checking whether a certificate record exists in the database. The user can enter a Certificate ID through the verification page. The system searches the database using the entered Certificate ID and the corresponding administrator information. If a matching certificate record is available, the system retrieves and displays important details such as Certificate ID, student name, register number, course, institution, and issue date. If no matching record is found, the system displays an appropriate message indicating that the certificate could not be found. The same verification concept is also connected with the QR code generated for each certificate. By scanning the QR code, the user can directly access the certificate details page without manually entering the Certificate ID. This module provides a simple and effective method for checking certificate records stored in the system.

5.10 CERTIFICATE SEARCH MODULE

The Certificate Search Module allows the administrator to quickly locate a particular certificate from the stored certificate records. As the number of certificates increases, manually searching through every certificate record becomes time-consuming. Therefore, the system provides a search field in the Manage Certificates page. The administrator can enter relevant information such as Certificate ID, student name, or register number. The system processes the search keyword and retrieves matching records from the SQLite database. The results are then displayed in an organized table containing the certificate information and available management options. The search function operates only on the certificate records associated with the currently logged-in administrator. This helps maintain proper separation of certificate records. The module improves the efficiency of certificate management by allowing administrators to locate records quickly without browsing through the complete certificate list.

5.11 CERTIFICATE MANAGEMENT MODULE

The Certificate Management Module provides the administrator with an organized interface for viewing and managing issued certificate records. All available certificates are displayed in a structured table containing information such as display ID, Certificate ID, student name, register number, course, issue date, QR code, download option, and management action. The administrator can use the search function to locate specific certificate records. The module also provides an option to remove an unwanted certificate record when required. When a certificate is deleted, its corresponding uploaded file and QR code image can also be removed from the respective storage folders, while the related database record is removed from the certificate table. The displayed certificate numbers are arranged sequentially according to the current record order. This module therefore provides a centralized interface for managing certificate records and maintaining an organized certificate database.

5.12 CERTIFICATE DOWNLOAD MODULE

The Certificate Download Module allows the administrator to download the original certificate file stored in the system. Each certificate record contains the name of its corresponding uploaded file, which is stored in the certificate upload folder. When the administrator selects the Download option from the Manage Certificates page or certificate details page, the system identifies the selected certificate using the administrator ID and Certificate ID. It then retrieves the corresponding file name from the database and locates the file in the upload storage folder. If the certificate record exists and the file is available, the system sends the file to the user's browser as a downloadable attachment. If the requested certificate cannot be found, the system displays an appropriate error message. This module provides convenient access to previously uploaded certificate files and avoids the need to search manually through the storage directory. It also connects the stored digital certificate record with its original uploaded document.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 HOME PAGE

The Home Page is the initial interface displayed when the system is accessed. It provides a simple and user-friendly entry point to the certificate verification system. The page contains the system title and navigation options such as Home and Admin Login. The administrator can access the system through the Admin Login option to issue, manage, and verify academic certificates. The interface is designed with a clear layout so that users can easily understand the purpose of the system and navigate to the required functionality.

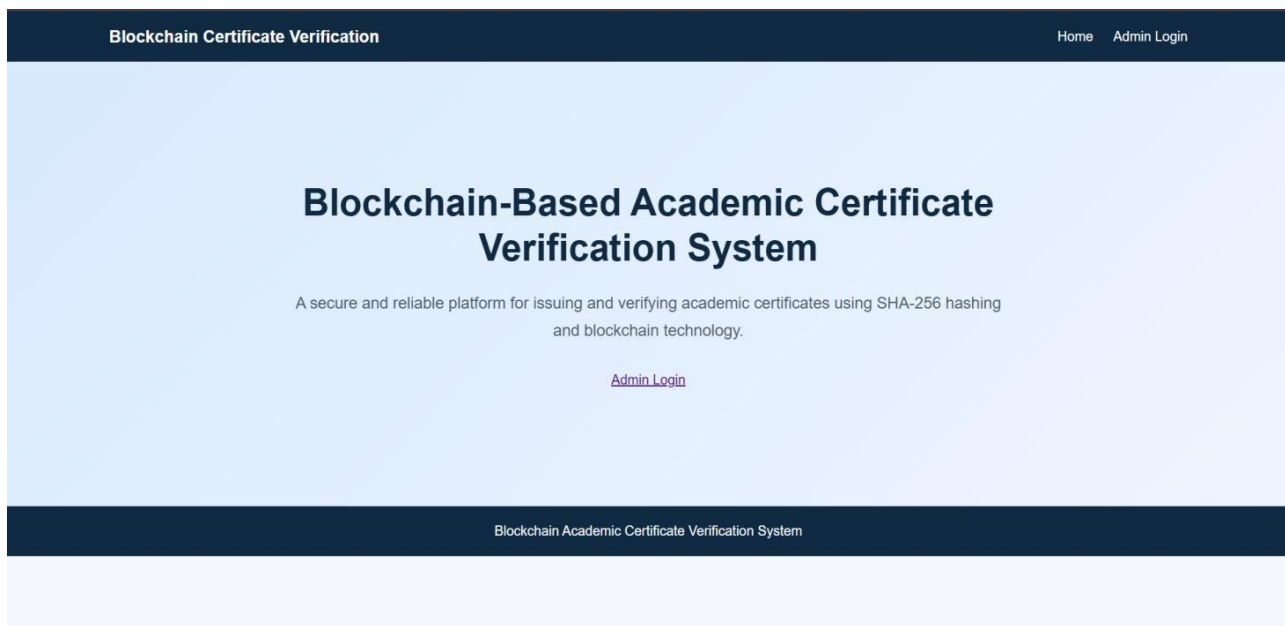


Figure 6.1 Home Page

6.2 ADMIN LOGIN PAGE

The Admin Login Page provides secure access to the administrative functions of the certificate verification system. The administrator must enter a valid username and password to access the system dashboard. After successful authentication, the administrator is directed to the dashboard, where certificate-related operations such as

issuing certificates, verifying certificates, managing certificate records, and viewing blockchain records can be performed. The login mechanism prevents unauthorized users from accessing the administrative functions and helps maintain the security of certificate records.

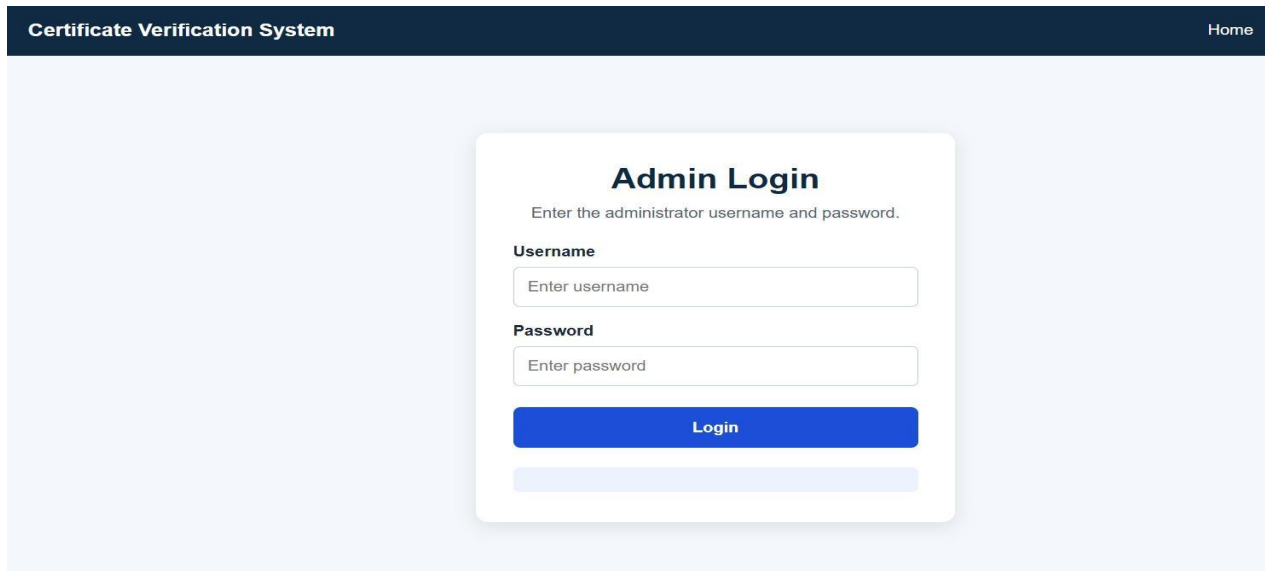
The image shows a web application interface for a 'Certificate Verification System'. At the top, there is a dark blue header bar with the text 'Certificate Verification System' on the left and a 'Home' link on the right. The main content area has a light blue background. In the center, there is a white card with a subtle shadow. The card is titled 'Admin Login' in bold black text. Below the title, it says 'Enter the administrator username and password.' in a smaller font. There are two input fields: the first is labeled 'Username' and contains the placeholder text 'Enter username'; the second is labeled 'Password' and contains the placeholder text 'Enter password'. Below these fields is a prominent blue button with the text 'Login' in white. At the bottom of the card, there is a thin, light blue horizontal bar.

Figure 6.2 Admin Login Page

6.3 DASHBOARD

The Dashboard is the main administrative interface of the certificate verification system. After successful login, the administrator is directed to the dashboard, which provides access to the major functions of the system. It includes options for issuing certificates, verifying certificates, viewing blockchain records, and managing issued certificates. The dashboard provides a centralized interface for performing certificate-related operations efficiently. It also helps the administrator navigate between different modules without requiring direct access to the underlying system components. The simple card-based layout improves usability and makes the available functions easy to identify.

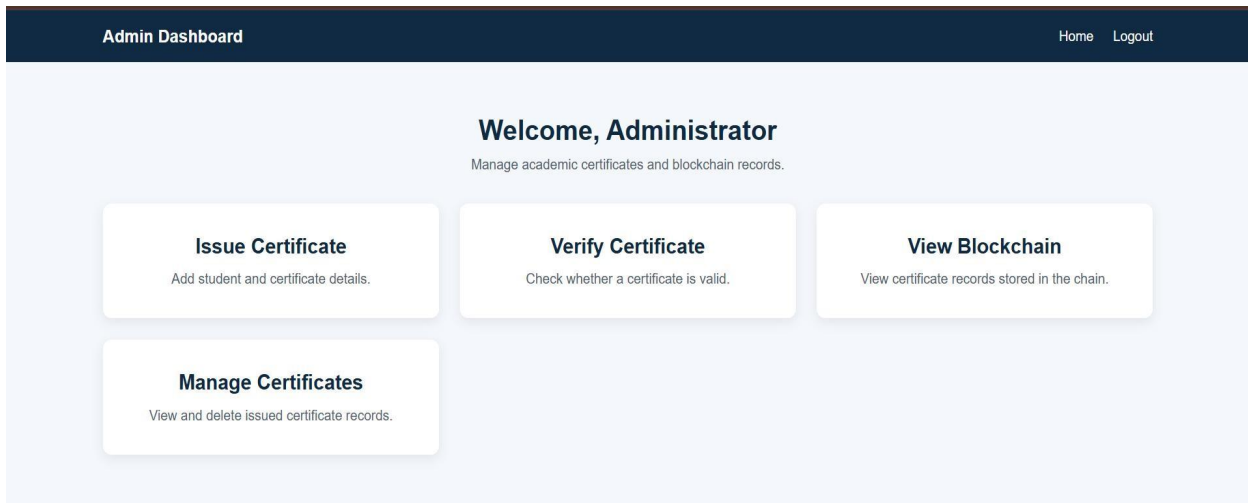


Figure 6.3 Dashboard

6.4 CERTIFICATE UPLOAD RESULT

The Certificate Upload module allows the administrator to enter the required academic certificate details and upload the corresponding certificate file into the system. The administrator provides information such as Certificate ID, Student Name, Register Number, Course, Institution, and Issue Date, and then selects the certificate file for uploading. After submission, the system processes the certificate information and stores the certificate record along with its associated file. The uploaded certificate is then available for further verification, management, QR code generation, and blockchain record creation. This module provides a structured method for registering academic certificates in the system.

Issue Academic Certificate

DashboardLogout

Issue Certificate

Enter the academic certificate details and upload the certificate.

Certificate ID

Example: KIOT-MECH-2026-001

Student Name

Enter student name

Register Number

Enter register number

Course

Example: B.E. Mechanical Engineering

Institution

Enter institution name

Issue Date

dd-mm-yyyy

Certificate File

Choose File | No file chosen

Issue Certificate

Figure 6.4 Certificate Upload

6.5 CERTIFICATE MANAGEMENT RESULT

The Certificate Management module provides the administrator with a centralized view of all certificates issued through the system. The records are displayed in a structured table containing the display ID, Certificate ID, student name, register number, course, issue date, QR code, download option, and action option. The administrator can search for a particular certificate and access the required certificate record easily. The displayed ID numbers are arranged sequentially according to the current certificate records, making the list easier to understand and manage. The module also provides options to access the stored certificate and perform necessary management operations. This result demonstrates that the system successfully stores and organizes certificate information in an accessible format.

Manage Certificates							
Dashboard Logout							
Issued Certificate Records							
Search Certificate						Search	
ID	Certificate ID	Student Name	Register Number	Course	Issue Date	Download	Action
1	123	gjhhd	65	mfmr	2005-11-15	Download	Delete
2	12	jjh	655	mech	52005-12-11	Download	Delete
3	mech12	lkk	655	mech	2005-11-15	Download	Delete
4	kl123	lok	566	lkk	2005-11-15	Download	Delete
5	me123	loi	6544	mjh	2005-11-15	Download	Delete
6	ml123	lko	654	mech	2002-11-15	Download	Delete
7	ab123	logesh	6112233	mech	0200-11-15	Download	Delete

Figure 6.5 Certificate Managemnt

6.6 CERTIFICATE SEARCH RESULT

The Certificate Search module enables the administrator to quickly locate a specific certificate from the stored certificate records. The administrator can enter a relevant keyword, such as the Certificate ID, Student Name, or Register Number, in the search field. The system processes the entered keyword and displays the matching certificate records in an organized format. This reduces the time required to manually identify a certificate from a large number of stored records. The search result displays the relevant certificate information and provides access to the available management functions. The module therefore improves the efficiency and usability of the certificate management system by providing a simple and effective method for retrieving required certificate records.

Manage Certificates							Dashboard	Logout
Issued Certificate Records								
log					Search			
ID	Certificate ID	Student Name	Register Number	Course	Issue Date	Download	Action	
1	ab123	logesh	6112233	mech	0200-11-15	Download	Delete	
2	test001	logesh	654789	mech	2005-11-15	Download	Delete	

Figure 6.6 Certificate Search

6.7 CERTIFICATE VERIFICATION RESULT

The Certificate Verification module is used to confirm the validity of an issued academic certificate using its Certificate ID. The administrator enters the required **Certificate ID**, and the system searches the stored certificate records for a matching entry. When the certificate is found, the system displays the corresponding certificate details, including the **student name, register number, course, institution, and issue date**. This allows the certificate information to be checked against the registered record in the system. If the entered **Certificate ID does not match** any stored record, the system indicates that the **certificate not found**. The verification result demonstrates that the system can efficiently retrieve and validate registered certificate information.

Verify Certificate

Certificate ID

Verify Certificate

Certificate Verified Successfully

Certificate ID	certificate123
Student Name	Logeshwaran
Register Number	61122314050
Course	Mech
Institution	kvit
Issue Date	2026-08-26
Status	VALID

Certificate not found.

Figure 6.7 Certificate Verification

6.8 QR CODE GENERATION RESULT

The QR Code Generation module generates a unique QR code for each issued academic certificate. The generated QR code is associated with the corresponding certificate record and contains the verification information required to access the certificate details. After a certificate is successfully registered, the system generates the QR code along with the certificate information. When the certificate is verified successfully, the generated QR code is displayed along with the certificate details. The QR code can be scanned by users or students to access the certificate verification information without manually entering the Certificate ID. This provides a convenient and efficient method for certificate verification and improves the accessibility of the system.

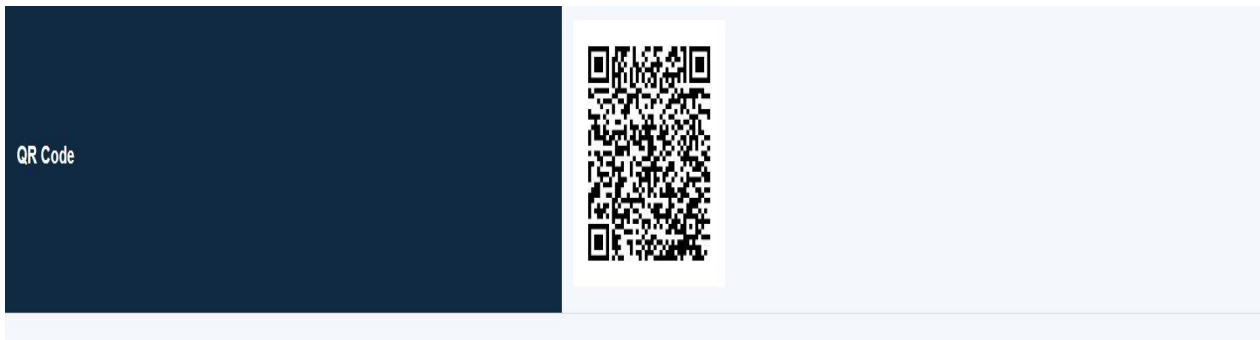


Figure 6.8 QR Code Generation

6.9 BLOCKCHAIN EXPLORER RESULT

The Blockchain Explorer module provides a visual representation of the academic certificate records stored in the blockchain structure. Each certificate record is displayed as an individual block containing important information such as Certificate ID, student name, register number, course, certificate status, previous hash, and block hash. The previous hash connects each block with the preceding block and establishes the chain structure. The Blockchain Explorer helps the administrator inspect the stored certificate records and understand how the certificate information is linked within the blockchain. This provides transparency and makes the stored records easier to examine.

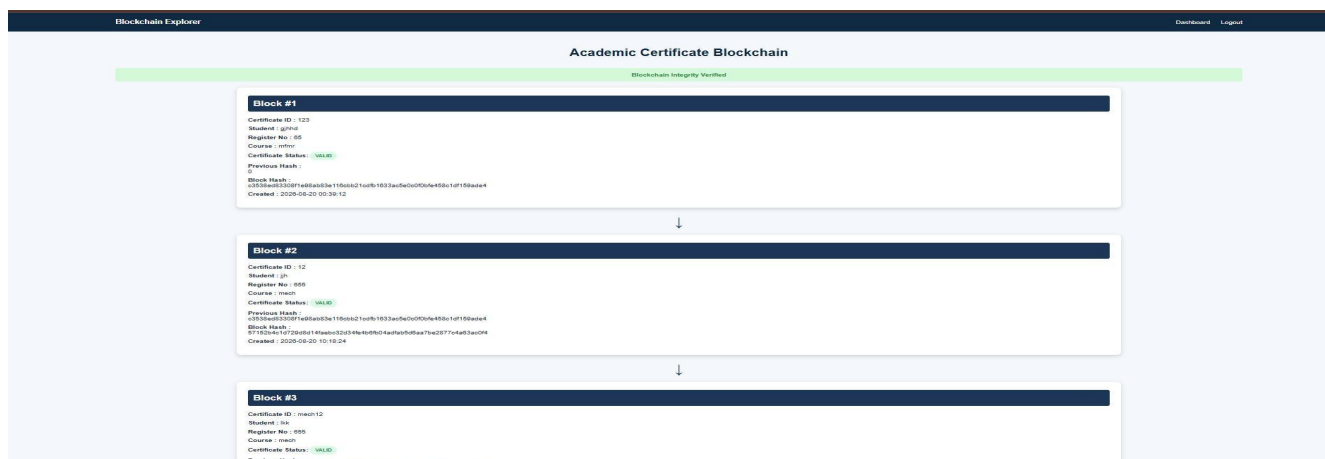


Figure 6.9 Blockchain Explorer

6.10 BLOCKCHAIN INTEGRITY RESULT

The Blockchain Integrity module verifies whether the stored blockchain records remain consistent and unchanged. The system checks the relationship between the

current block and the previous block by comparing the stored hash values. If the hash relationships are valid, the system displays an integrity verification status indicating that the blockchain records are valid. This process helps identify unauthorized modification or inconsistency in the stored certificate records. The integrity verification feature therefore provides an additional layer of reliability and helps maintain the trustworthiness of certificate information stored in the system.

“The blockchain integrity verification result is shown in Figure 6.9, where the system displays ‘Blockchain Integrity Verified’ after successfully validating the hash relationship between the stored blocks.”

6.11 CERTIFICATE DOWNLOAD RESULT

The Certificate Download module allows the administrator or authorized user to download the original certificate associated with a verified certificate record. After successful certificate verification, the system provides a Download Certificate option. Selecting this option retrieves the corresponding certificate file from the system's storage location and makes it available to the user. This feature provides convenient access to the original certificate without requiring the certificate to be uploaded again. It also confirms that the certificate record stored in the system is linked with its corresponding certificate file.

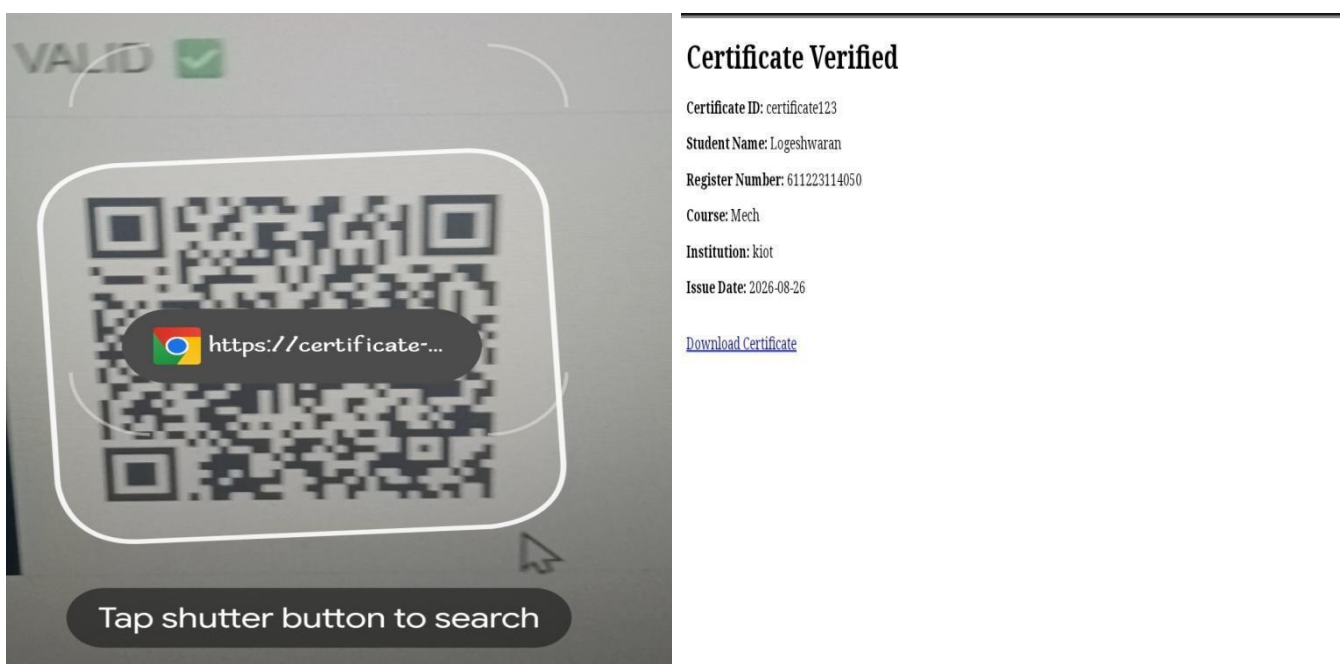


figure 6.10 Certificate Download

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Blockchain-Based Academic Certificate Verification System was developed to provide a secure, reliable, and efficient solution for registering and verifying academic certificates. The proposed system addresses the limitations of traditional certificate verification methods, which can involve manual checking, time consumption, and difficulties in identifying fraudulent certificates.

The developed system integrates a web-based application with a database, SHA-256 hashing, blockchain-based record linking, and QR code technology. Administrators can log in to the system, upload certificate information, manage registered certificates, search for certificates, and verify certificate details. Each registered certificate is associated with a blockchain record containing the previous hash and block hash, which helps maintain the integrity of the stored records.

The certificate verification module enables users to verify certificates using the Certificate ID. In addition, the QR code provides a convenient method for accessing certificate verification information without manually entering the Certificate ID. The Blockchain Explorer provides a visual representation of the certificate records and their hash relationships, while the blockchain integrity verification feature helps confirm that the stored records have not been altered.

The certificate download feature further improves the usability of the system by allowing verified certificate information to be accessed conveniently. Testing and validation of the major modules demonstrated that the implemented system performs the required certificate registration, management, search, verification, QR code generation, blockchain record creation, integrity checking, and certificate download functions successfully.

Overall, the developed prototype demonstrates that blockchain technology can be effectively applied to academic certificate verification. The system provides a transparent and efficient approach for maintaining certificate records and reducing the effort required for certificate authentication. Therefore, the proposed system

successfully achieves its primary objective of developing a digital platform for secure and convenient academic certificate verification.

7.2 FUTURE SCOPE

The developed system can be further enhanced with additional features and technologies to make it suitable for large-scale real-world implementation. The following improvements can be considered as future developments.

1. Decentralized Blockchain Network

The current prototype can be extended from a local blockchain implementation to a decentralized blockchain network. Multiple educational institutions can participate in the network and maintain distributed copies of certificate records.

2. Smart Contract Integration

Smart contracts can be incorporated to automate certificate registration and verification processes. A smart contract can validate certificate information and control the creation and verification of certificate records without depending entirely on manual administrative operations.

3. Multi-Institution Support

The system can be extended to support multiple colleges, universities, and educational institutions. Each institution can have its own administrative account and securely manage certificates issued by the institution.

4. Mobile Application

A dedicated Android or cross-platform mobile application can be developed for certificate verification. Users can scan the QR code using their mobile device and immediately view the verification status and certificate details.

5. Enhanced Security

Additional security mechanisms such as multi-factor authentication, role-based access control, encrypted communication, secure password hashing, and digital signatures can be implemented to improve the overall security of the system.

6. Cloud Deployment

The system can be deployed on a reliable cloud platform to provide continuous access to certificate verification services. Cloud deployment can also improve scalability, availability, backup, and data management.

7. Integration with Institutional Databases

The proposed system can be integrated with existing college and university academic databases. This would allow certificate information to be automatically retrieved from institutional records and reduce duplicate data entry.

8. Advanced Fraud Detection

Future versions can include advanced mechanisms for detecting altered or fraudulent certificates. Digital signatures, document fingerprinting, advanced hash comparison, and intelligent document analysis can be incorporated to improve authenticity detection.

9. Public Verification Portal

A public verification portal can be developed where employers, government organizations, and other authorized parties can verify certificates without requiring direct access to institutional administrative systems.

10. Improved Blockchain Analytics

The Blockchain Explorer can be enhanced with advanced visualization features such as block timelines, transaction information, certificate status history, hash comparison, and detailed integrity reports. This would make blockchain records easier to understand and audit.

CHAPTER 8

APPENDICES

8.1 PSEUDOCODE

Algorithm 1: Administrator Login

```
BEGIN  
  
Display Login Page  
  
Input Username and Password  
  
IF Username and Password are correct THEN  
    Create Administrator Session  
    Open Dashboard  
ELSE  
    Display "Invalid Login Credentials"  
END IF  
  
END
```

Algorithm 2: Certificate Upload

```
BEGIN  
  
Login as Administrator  
  
Display Certificate Upload Page  
  
Enter:  
    Certificate ID  
    Student Name  
    Register Number  
    Course  
    Institution  
    Issue Date  
  
Select Certificate File  
  
IF File Format is Valid THEN  
    Store Uploaded Certificate File  
    Generate SHA-256 Hash  
    Retrieve Previous Block Hash  
    Generate Current Block Hash  
    Generate QR Code
```

```
    Store Certificate Details in Database
    Display "Certificate Uploaded Successfully"
ELSE
    Display "Invalid File Format"
END IF

END
```

Algorithm 3: SHA-256 Hash Generation

```
BEGIN

Read Uploaded Certificate File

Convert Certificate File into Binary Data

Apply SHA-256 Hashing Algorithm

Generate 64-Character Hash Value

Return Certificate Hash

END
```

Algorithm 4: Blockchain Record Creation

```
BEGIN

Receive Certificate Details

Retrieve Previous Block Hash

IF No Previous Block Exists THEN
    Set Previous Hash = "0"
ELSE
    Retrieve Hash of Previous Block
END IF

Generate Current Timestamp

Combine:
    Certificate ID
    Certificate Hash
    Previous Hash
    Timestamp
```

```
Generate Current Block Hash  
Store Blockchain Record in Database  
END
```

Algorithm 5: QR Code Generation

```
BEGIN  
Receive Certificate ID  
Create Certificate Verification URL  
Generate QR Code from Verification URL  
Save QR Code Image  
Store QR Code Path in Database  
Associate QR Code with Certificate Record  
END
```

Algorithm 6: Certificate Verification

```
BEGIN  
Open Certificate Verification Page  
Enter Certificate ID  
Search Certificate ID in Database  
IF Certificate ID is Not Found THEN  
    Display "Certificate Not Found"  
ELSE  
    Retrieve Certificate Details  
    Retrieve Stored Certificate Hash  
    Display Certificate Verification Result  
    Display Certificate Status as "VALID"  
END IF  
END
```


Algorithm 7: Blockchain Integrity Verification

BEGIN

Retrieve All Blockchain Records

Set Previous Hash = "0"

FOR Each Blockchain Block

 Recalculate Current Block Hash

 Compare Stored Hash with Recalculated Hash

 Compare Current Previous Hash with Previous Block Hash

 IF Hash Mismatch THEN

 Display "Blockchain Integrity Invalid"

 STOP

 END IF

 Update Previous Hash

END FOR

Display "Blockchain Integrity Verified"

END

Algorithm 8: Certificate Search

BEGIN

Open Certificate Search Page

Enter Search Keyword

Search Database Using:

 Certificate ID

 Student Name

 Register Number

Retrieve Matching Certificate Records

IF Matching Records are Found THEN

```
    Display Certificate Records
ELSE
    Display "No Certificates Found"
END IF

END
```

Algorithm 9: Certificate Download

```
BEGIN

Select Certificate

Retrieve Certificate File Path

IF Certificate File Exists THEN
    Provide Download Certificate Option
    Download Certificate File
ELSE
    Display "File Not Found"
END IF

END
```

Algorithm 10: Blockchain Explorer

```
BEGIN

Open Blockchain Explorer Page

Retrieve All Blockchain Records

FOR Each Blockchain Block

    Display Block Number

    Display Certificate ID

    Display Student Name

    Display Register Number

    Display Course

    Display Certificate Status
```

```
Display Previous Hash  
Display Current Block Hash  
Display Created Timestamp  
END FOR  
Display Blockchain Integrity Status  
END
```

Algorithm 11: Administrator Logout

```
BEGIN  
Clear Administrator Session  
Destroy Login Session  
Redirect to Home Page  
END
```

8.2 HTML TEMPLATES

The frontend of the project was developed using HTML templates with the Jinja2 templating engine supported by Flask. The templates provide the user interface for certificate registration, management, verification, searching, blockchain record viewing, and certificate downloading.

Templates Included

- index.html – Home Page
- login.html – Administrator Login Page
- dashboard.html – Administrator Dashboard
- upload.html – Certificate Upload Page
- verify.html – Certificate Verification Page
- result.html – Certificate Verification Result
- certificates.html – Certificate Management Page
- blockchain.html – Blockchain Explorer
- certificate_details.html – Certificate Details and Verification Page

The HTML templates are connected with the Flask application and use Jinja2 expressions to dynamically display certificate information retrieved from the SQLite database. The templates provide a simple and user-friendly interface for performing certificate-related operations.

8.3 CSS STYLESHEET

The appearance and layout of the web application are designed using a CSS stylesheet.

File:

style.css

The stylesheet provides styling for the major components of the application, including:

- Responsive page layout
- Navigation bar
- Home page
- Login form
- Dashboard cards
- Certificate upload form
- Certificate verification form
- Search interface
- Certificate tables
- Buttons and input controls
- Verification status messages
- QR code display
- Blockchain Explorer interface
- Certificate details page
- Download certificate interface

The CSS stylesheet maintains a consistent visual appearance across the different pages of the certificate verification system and improves the overall usability of the application.

8.4 DATABASE SCHEMA

The application uses SQLite as the database management system for storing administrator and academic certificate information.

Database Name:

database.db

The **certificates table** stores the complete information associated with each registered academic certificate, including certificate details, uploaded file information, SHA-256 hash, blockchain hash values, and QR code information.

Field Name	Data Type	Description
id	INTEGER	Primary key and unique record number
admin_id	INTEGER	Identifies the administrator associated with the certificate
certificate_id	TEXT	Unique certificate identification number
student_name	TEXT	Name of the student
register_number	TEXT	Student register number
course	TEXT	Name of the course
institution	TEXT	Name of the educational institution
issue_date	TEXT	Date on which the certificate was issued
file_name	TEXT	Name of the uploaded certificate file
qr_code	TEXT	QR code image file name
certificate_hash	TEXT	SHA-256 hash generated from the certificate file
previous_hash	TEXT	Hash value of the previous blockchain record
block_hash	TEXT	Hash value of the current blockchain record
created_at	TEXT	Date and time when the certificate record was created
status	TEXT	Current certificate status, such as VALID

Table 8.1 Certificate Database Schema

The **admins table** stores the login credentials and institution information required for administrator authentication.

Field Name	Data Type	Description
admin_id	INTEGER	Primary key and unique administrator ID
college_name	TEXT	Name of the institution associated with the administrator
username	TEXT	Administrator login username
password	TEXT	Administrator login password

Table 8.2 Admin Database Schema

The database establishes a relationship between the administrator and the certificates through the admin_id field. This allows certificate records to be associated with the corresponding administrator account. A unique index is also maintained on admin_id and certificate_id to prevent duplicate certificate IDs under the same administrator.

CHAPTER 9

REFERENCES

- 1) S. Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System, 2008.
- 2) M. Crosby, P. Pattanayak, S. Verma, and V. Kalyanaraman, “Blockchain Technology: Beyond Bitcoin,” *Applied Innovation Review*, no. 2, pp. 6-19, Jun. 2016.
- 3) M. Swan, *Blockchain: Blueprint for a New Economy*. Sebastopol, CA, USA: O’Reilly Media, 2015.
- 4) W. Diffie and M. E. Hellman, “New Directions in Cryptography,” *IEEE Transactions on Information Theory*, vol. 22, no. 6, pp. 644-654, Nov. 1976.
- 5) National Institute of Standards and Technology (NIST), *Secure Hash Standard (SHS)*, FIPS PUB 180-4, 2015.
- 6) A. Narayanan, J. Bonneau, E. Felten, A. Miller, and S. Goldfeder, *Bitcoin and Cryptocurrency Technologies*. Princeton University Press, 2016.
- 7) D. Yaga, P. Mell, N. Roby, and K. Scarfone, *Blockchain Technology Overview*, NIST Special Publication 800-193, 2018.
- 8) M. Pilkington, “Blockchain Technology: Principles and Applications,” in *Research Handbook on Digital Transformations*, Edward Elgar Publishing, 2016.
- 9) C. Dwork and M. Naor, “Pricing via Processing or Combatting Junk Mail,” in *Advances in Cryptology – CRYPTO’92*, Springer, pp. 139-147, 1993.
- 10) M. Herlihy, “Blockchains and the Future of Distributed Computing,” *Communications of the ACM*, vol. 62, no. 2, pp. 78-85, Feb. 2019.
- 11) A. M. Antonopoulos, *Mastering Bitcoin*, 2nd ed. Sebastopol, CA: O’Reilly Media, 2017.
- 12) G. Wood, *Ethereum: A Secure Decentralised Generalised Transaction Ledger*, Ethereum Foundation, Yellow Paper, 2014.
- 13) Hyperledger Foundation, *Hyperledger Fabric Documentation*, Linux Foundation, 2024.

- 14)** M. Grinberg, Flask Web Development, 2nd ed. Sebastopol, CA: O'Reilly Media, 2018.
- 15)** R. Hettinger, "Python Programming Language Documentation," Python Software Foundation, 2024.
- 16)** SQLite Consortium, SQLite Documentation, SQLite Official Documentation, 2024.
- 17)** T. Denso Wave Incorporated, QR Code Essentials and Specifications, DENSO WAVE, Japan.
- 18)** R. Rivest, "The MD5 Message-Digest Algorithm," RFC 1321, Internet Engineering Task Force (IETF), Apr. 1992.
- 19)** M. Bellare and P. Rogaway, "Introduction to Modern Cryptography," University of California, Davis, Lecture Notes, 2005.
- 20)** T. Berners-Lee, "Information Management: A Proposal," CERN, Geneva, Switzerland, 1989.
- 21)** World Wide Web Consortium (W3C), HTML5: A Vocabulary and Associated APIs for HTML and XHTML , W3C Recommendation, 2021.
- 22)** World Wide Web Consortium (W3C), Cascading Style Sheets (CSS) Level 3 Specification, W3C Recommendation, 2021.
- 23)** Ecma International, ECMAScript(R) 2023 Language Specification, ECMA-262, 14th Edition, 2023.
- 24)** A. Silberschatz, H. F. Korth, and S. Sudarshan, Database System Concepts, 7th ed. New York, NY, USA: McGraw-Hill Education, 2020.
- 25)** I. Sommerville, Software Engineering, 10th ed. Boston, MA, USA: Pearson Education, 2016.